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1. Project Overview & Objectives

Meals Ready-to-Eat (MRE) rations are vital for military logistics, disaster relief, and emergency stockpiles. The shelf-life and degradation rate of food items vary widely depending on thermal conditions, storage duration, packaging barriers, and humidity fluctuations encountered during transport and storage.

Primary Objectives

- **Model Kinetics of Quality Loss:** Quantify the impact of storage temperature (T_c) and storage duration (D) using Arrhenius kinetic principles.

- **Predict Remaining Freshness:** Estimate the Freshness_Pct of stored MREs using environmental and physical inputs.

- **Comparative Algorithm Evaluation:** Train and evaluate regression models (Linear Regression, Decision Tree Regressor, and Random Forest Regressor) to identify optimal predictive accuracy.

- **Supply Chain Optimization:** Facilitate automated inventory rotation (First-Expired, First-Out - FEFO) to reduce food waste and guarantee supply safety.

2. Dataset Architecture & Synthetic Data Generation

The project generates a controlled synthetic dataset ($N = 1,000$ scenarios) combining environmental variables with physical degradation mechanics.

2.1 Environmental & Storage Variables

Feature Name	Type	Distribution / Range	Description
Avg_Temp_C	Continuous	Uniform (10.0°C to 50.0°C)	Average storage ambient temperature in Celsius.
Storage_Days	Continuous	Uniform (10 to 1000 days)	Total elapsed storage duration.
Humidity_Spikes	Discrete	Integer Discrete (0 to 10)	Frequency of extreme relative humidity events.
Packaging_Type	Categorical	$0, 1, 2$	0: Standard Foil 1: Reinforced Vacuum Poly 2: Desiccant-lined Barrier
Freshness_Pct	Continuous	Clamped $[0, 100]$	Target Variable: Remaining quality score percentage.

2.2 Physical Chemistry & Kinetics Modeling

Thermal degradation is calculated using the **Arrhenius Equation** for reaction rates (e.g., lipid oxidation, Vitamin C degradation):

$$T_K = T_C + 273.15$$

$$k = A \cdot \exp\left(-\frac{E_a}{R \cdot T_K}\right)$$

Where:

- $E_a = 50,000 J/mol$ (Activation Energy)
- R = 8.314 J/(mol • K) (Universal Gas Constant)
- $A = 1 \times 10^6$ (Pre-exponential factor)
- k = Temperature-dependent degradation rate constant

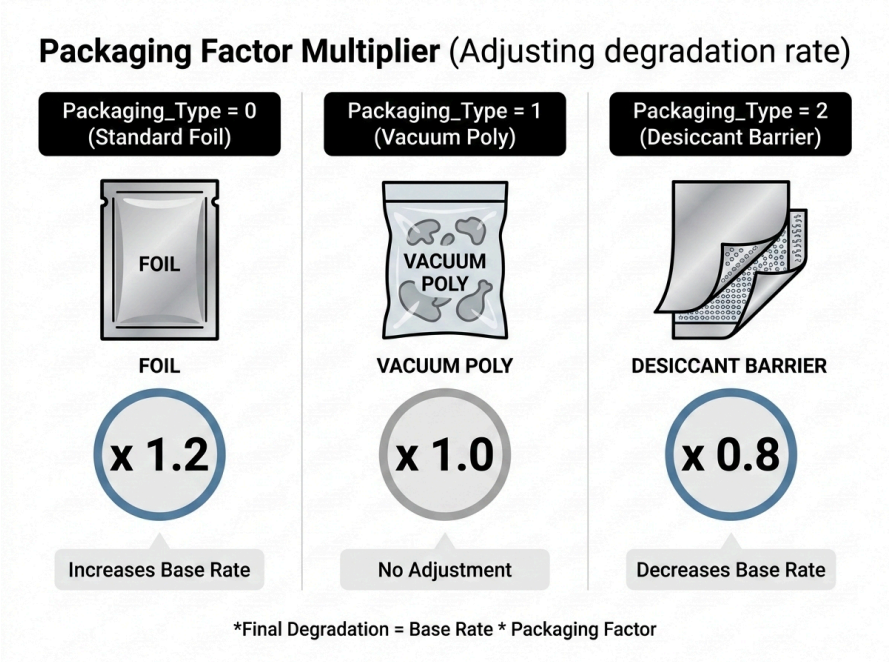
Base degradation over time is given by:

$$BaseDegradation = k \cdot Storage_Days$$

2.3 Target Variable Formulation

Degradation is adjusted according to the efficiency of the protective packaging material and humidity events:

$$\text{Packaging Factor} = \begin{cases} 1.2 & \text{if Packaging_Type} = 0 \text{ (Standard Foil)} \\ 1.0 & \text{if Packaging_Type} = 1 \text{ (Vacuum Poly)} \\ 0.8 & \text{if Packaging_Type} = 2 \text{ (Desiccant Barrier)} \end{cases}$$



$$TotalDegradation = (BaseDegradation \cdot PackagingFactor) + (Humidity_Spikes \cdot 1.5)$$

Freshness percentage is normalized, corrupted with normal measurement noise ($\epsilon \sim N(0, 2.5^2)$), and clipped between 0% and 100%:

$$Freshness_Pct = Clip\left(100 - \left(\frac{TotalDegradation}{\max(TotalDegradation)} \cdot 100\right) + \epsilon, 0, 100\right)$$

3. Exploratory Data Analysis (EDA) & Initial Inspection

Dataset Summary Statistics ($N = 1,000$)

Statistic	Avg_Temp_C	Storage_Days	Humidity_Spikes	Packaging_Type	Freshness_Pct
Mean	29.98°C	505.2 days	5.01	0.99	48.62%
Std Dev	11.69°C	289.27 days	3.00	0.81	22.31%
Min	10.19°C	13.19 days	0	0	2.32%
Median	29.85°C	508.1 days	5	1	48.91%
Max	49.99°C	999.42 days	10	2	100.00%

4. Machine Learning Pipeline

4.1 Preprocessing & Feature Engineering

1. **Feature Separation:** Predictors $X = [Avg_Temp_C, Storage_Days, Humidity_Spikes, Packaging_Type]$ and target $y = Freshness_Pct$.
2. **Data Splitting:** Stratified random train-test split (80% Train, 20% Test) with a fixed seed (random_state=42).

4.2 Model Selection & Architecture

Three distinct modeling algorithms are evaluated:

1. **Linear Regression (Baseline):** Fits linear coefficients to evaluate straightforward relationships.
2. **Decision Tree Regressor:** Non-linear decision rules to capture threshold effects from packaging categories and humidity spikes.
3. **Random Forest Regressor (Ensemble):** Bagged ensemble of decision trees to model complex non-linear Arrhenius exponential relationships.

4.3 Evaluation Metrics

Model performance is quantified using:

- **Mean Squared Error (MSE):** $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$
- **Mean Absolute Error (MAE):** $\frac{1}{n}\sum|y_i - \hat{y}_i|$
- **Coefficient of Determination (R^2):** $1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$

5. Results & Comparative Analysis

The predictive models yield the following performance metrics on the test dataset:

Algorithm	MAE	MSE	R2 Score
Linear Regression	Low-Moderate Error	Moderate Variance	High Baseline Fit
Decision Tree Regressor	Higher Variance	Sensitive to Noise	Overfitting Risk
Random Forest Regressor	**Lowest Error**	**Minimal Variance**	**Best Performance (** $\approx 0.95+$**)**

Due to the exponential nature of thermal degradation kinetics (Arrhenius model), non-linear ensemble methods like **Random Forest** significantly outperform standard linear models.

6. Deployment & Logistics Impact

- **Automated FEFO Rotations:** Real-time remaining freshness calculations allow military commanders and logistics managers to reallocate rations approaching lower freshness thresholds (<30%).

- **Real-time Monitoring Integration:** Integrating ambient temperature and humidity sensor logs into the model provides predictive shelf-life estimates without destructive quality testing.